The 15:50 Burst: Minute-Level Price Pressure around MSCI U.S. Index Reconstitutions

Point-in-Time Event Construction, Review-Date Inference, and an Exploratory Top-10 Portfolio

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Abstract

This paper studies direction-signed minute-level price pressure around MSCI U.S. Standard Index implementation closes. The calendar contains 27 scheduled reviews, of which 26 have nonempty U.S. Standard changes from May 2020 through August 2026. The analyzed sample has 463 public addition/deletion events and 459 usable 15:49 minute windows. The baseline proxy enters at the open of the start-stamped 15:49 bar and exits at the close of the start-stamped 15:50 bar, approximately 15:50:59. At the user-specified 0 bp transaction cost, the all-name direction-signed basket earns +13.54 bp per scheduled review, compounds to +3.57%, and has an event Sharpe of 1.28. The interval was selected from 29 adjacent windows; its four-review block max-t p-value is 0.164, and performance is concentrated after 2023. An exploratory balanced top-10 construction earns +12.58 bp per scheduled review at 0 bp, with +3.32% cumulative return, 0.52% cash-funded CAGR, 2.00 active-trade Sharpe, and -0.09% event-step maximum drawdown. Its gross lift over the same-gate capped basket does not survive the 15-rule family adjustment (block max-t p=0.352). Additions contribute 59.1% and deletion shorts 40.9% of arithmetic gross profit. In 2020-2022, however, additions earn 72.89 bp while deletion shorts lose -9.95 bp; this leg cancellation is the principal accounting source of the weak early estimate. A pre-15:49 migration-aware gate raises the 10 bp active-date hit rate from 70.6% to 87.5%, but retains only 75.7% of baseline arithmetic net profit and does not improve paired return after family adjustment. The results document a research effect, not executable arbitrage: prices are minute OHLCV proxies, the rank omits licensed MSCI flows and float-adjusted capitalization, and the test has no historical NBBO, auction-imbalance, depth, latency, borrow, or option-quote replay.

Keywords: index reconstitution, closing auction, price pressure, event study, point-in-time data, block bootstrap, multiple testing, portfolio construction.

1 Introduction

Index-tracking funds are evaluated against closing benchmark weights. When an index provider changes constituents or float-adjusted weights, benchmark-aware portfolios have an incentive to complete residual trading near the implementation close. Prior work documents announcement-to-implementation effects and unusually large closing-auction activity around MSCI reconstitutions [1]. Separate market-microstructure research shows that closing-imbalance dissemination can move prices before the auction itself [2-5]. The research question is whether these mechanisms produce a repeatable, direction-signed burst in U.S. stocks during the minute surrounding 15:50 Eastern Time.

The empirical problem is unusually easy to overstate. Hundreds of stocks trade on each review date, but the independent shock is the review date, not the security. The exact interval was identified after inspecting the event-time profile. Public addition/deletion lists omit weight-only changes and do not reveal net product-level forced flow. Finally, a minute-bar open or close is not an executable bid or offer. The paper therefore treats review dates as the statistical units, adjusts for window search, keeps the 0 bp user scenario separate from a 10 bp execution sensitivity, and labels every fill as a research proxy.

The paper makes five contributions. First, it constructs a point-in-time event panel from official public MSCI review material. Second, it decomposes the closing window at one-minute frequency and tests the 15:49-open to 15:50-bar-close interval. Third, it evaluates a pre-event top-10 concentration rule with explicit cash gates, weight caps, event-step performance, and security-level attribution. Fourth, it tests pre-entry stock, market, liquidity, volatility, and migration-state filters under a 15:47 safe cutoff. Fifth, it specifies the message-level data and sequential holdout evidence required before the strategy can be treated as deployable.

2 Institutional Setting and Hypotheses

2.1 Reconstitution and the closing auction

MSCI announces periodic reviews before implementation. Index funds, separately managed accounts, derivatives desks, and benchmark-aware active managers may trade in advance, in the continuous market, or in the official closing auction. Their aggregate residual demand depends on old and new index weights, product assets, replication choices, crossing opportunities, and overlap across Standard, Small Cap, IMI, World, and ACWI families. An addition to Standard can simultaneously be a deletion from Small Cap, so a binary label is not an exact forced-flow sign.

In the U.S. closing process, venue-specific imbalance information becomes available before the 16:00 auction. Nasdaq historically begins early imbalance dissemination around 15:50, while current NYSE auction procedures also publish imbalance information during the final minutes [2-4, 7-8]. Fast participants can incorporate indicative demand into continuous-market prices before the official close. This motivates a narrow event window without establishing that the observed minute bar is caused by a particular imbalance message.

2.2 Mechanism and testable predictions

$$ExpectedFlow_{i,t} = sum_{p}\, AUM_{p,t}\, replication_{p,t}\, (weight_{i,t}^{post} - weight_{i,t}^{pre}) \quad (1)$$

Equation (1) defines the production quantity that should drive the trade; related benchmark-intensity evidence links assets times benchmark weights to price pressure [25]. It cannot be reconstructed from public lists alone. The current study instead tests the reduced-form implication that additions tend to move up and deletions down during the disclosure interval. The primary hypothesis is a positive mean direction-signed return from the 15:49 bar open to the 15:50 bar close. The second hypothesis is that the event-date effect exceeds the same-name, same-clock return from nearby prior sessions. The third is that a pre-event liquidity/volatility rank can concentrate the effect without creating excessive name risk.

3 Data, Timing, and Event Construction

3.1 Sample and source lineage

Table 1: Sample construction and measurement coverage.

Item	Count	Definition
Scheduled / nonempty dates	27 / 26	One empty February 2020 review; analyzed sample begins May 2020
Public security events	463	U.S. Standard additions and deletions after ticker resolution
Usable 15:49 minute windows	459	99.1% of public events
Usable daily-close proxies	451	97.4% of public events
Controlled review dates	24	Four complete prior sessions available
Controlled security-events	398	Same-name pre-event comparison panel
Top-10 portfolio dates	26	17 active, 9 cash
Top-10 security records	161	80 long and 81 short research-proxy positions

Official public review lists identify security names and announced additions/deletions. Permanent identifiers and complete pro-forma index weights are unavailable in the public source. Ticker mappings are therefore resolved with an explicit override ledger, and coverage failures remain in the audit output rather than being silently dropped. MarketData minute OHLCV is used for the underlying price path [9]. Its timestamps identify the start of each candle. The 15:50 exit proxy is consequently the close of the bar labeled 15:50, approximately 15:50:59.

The base impact ranking is frozen before the event session. The second-stage state filter uses only bars completed by 15:47:59.999 Eastern Time, leaving one full minute before the 15:49 entry proxy; daily liquidity features use data no

later than the preceding session. No 15:48 bar, 15:50 imbalance, entry-window return, implementation-session daily close, or later observation is permitted to enter the production-safe rule.

4 Empirical Design and Portfolio Construction

4.1 Direction-signed event return

$$SignedReturn_{i,t} = side_{i,t} (Price_{i,t,15:50\text{-}close} / Price_{i,t,15:49\text{-}open} - 1), \text{ where side = +1 for additions and -1 for deletions} \quad (2)$$

$$ReviewReturn_t = 0.5\ mean_{adds}(SignedReturn_{i,t}) + 0.5\ mean_{deletes}(SignedReturn_{i,t}) \quad (3)$$

The date portfolio balances the addition and deletion legs so dates with many deletions do not mechanically dominate. If a leg is unavailable, its capital remains in cash. Security observations within a date are not treated as independent. The review return is the atomic observation for t-tests, sign tests, block bootstraps, equity curves, and multiple-testing adjustments.

4.2 Exploratory top-10 impact proxy

$$ImpactProxy_{i,t} = Volatility_{i,t,20d}\ sqrt(\ ADV_{i,t,20d} / ExpectedClockDollarVolume_{i,t}\ ) \quad (4)$$

Equation (4) is a shadow ranking score, not an estimate of MSCI capital. It favors securities with high pre-event volatility and low expected same-clock liquidity relative to a synthetic order-size proxy. On each eligible date, the strategy selects up to five additions and five deletions. Each name receives at most 10% of capital; each leg receives at most 50%. A date remains entirely in cash unless at least three point-in-time-valid names are available on both sides. This produces 17 active and nine cash reviews, with six to ten names on active dates.

Table 2: Strategy definitions and research status.

Rule	Signal / selection	Weights	Cost treatment	Role
All-name burst	Public add/delete sign; all usable names	50% per leg; date-balanced	0 bp primary; 10 bp sensitivity	Reduced-form baseline
Capped valid universe	All PIT-valid names after three-per-side gate	10% name cap; residual cash	0 and 10 bp	Matched-gate control
Impact top-10	Up to five per side by Equation (4)	10% name cap; up to 100% gross	0 bp primary; 10 bp sensitivity	Exploratory strategy
State-filter challenger	15:47 stock/SPY state plus migration gate	Original cap; rebalance legs down; residual cash	10 bp comparison	Retrospective risk control
Volatility diagnostic	Absolute 15:49-15:59 movement	No investable option weights	Underlying only	Mechanism test

4.3 Performance and inference

$$Wealth_T = product_{t=1...T}(1 + ReviewReturn_t) \quad (5)$$

$$ActiveTradeSharpe = sqrt(4)\ mean(ReviewReturn_t \mid GrossExposure_t > 0) / sd(ReviewReturn_t \mid GrossExposure_t > 0) \quad (6)$$

$$Drawdown_t = Wealth_t / max_{s<=t}(Wealth_s) - 1 \quad (7)$$

Active-trade Sharpe annualizes the conditional trade series by sqrt(4) and excludes every zero-exposure review. The raw output retains Scheduled Event Sharpe as a separately labeled coverage diagnostic. The current risk-free input is zero per one-minute event until a point-in-time funding series is joined. CAGR is cash-funded over elapsed calendar time, despite trading only on implementation dates. Four-review moving blocks preserve serial dependence. The time-window family contains 29 adjacent two-bar candidates; the portfolio family contains 15 combinations of ranker and top-5/top-10/top-20 breadth. These adjustments are essential because the reported interval and concentrated rule were selected after exploration [10].

5 Empirical Results

5.1 Headline scorecard

Table 3: Cash-funded event performance, May 2020-August 2026.

Portfolio	Cost	Mean / sched.	Cumulative	CAGR	Active Sharpe	Max DD	Block mean CI
All-name burst	0 bp	+13.54 bp	+3.57%	+0.56%	1.28	-0.65%	[+2.51, +24.51] bp
All-name burst	10 bp sensitivity	+3.73 bp	+0.97%	+0.15%	0.35	-1.28%	[-7.42, +14.72] bp
Capped full valid universe	0 bp	+10.32 bp	+2.71%	+0.43%	1.78	-0.22%	[+2.73, +19.59] bp
Impact top-10	0 bp primary	+12.58 bp	+3.32%	+0.52%	2.00	-0.09%	[+3.94, +23.92] bp
Impact top-10	10 bp sensitivity	+6.50 bp	+1.70%	+0.27%	1.05	-0.42%	[-1.44, +16.56] bp

Notes: The 0 bp rows implement the user's selected transaction-cost assumption. The 10 bp rows are execution sensitivities, not the primary scenario. Active Sharpe uses sqrt(4) and excludes zero-exposure review dates; Scheduled Event Sharpe remains in the downloadable metrics for coverage analysis. CAGR is cash-funded over elapsed calendar time; Max DD is event-step. Prices are minute-bar proxies rather than executable quotes.

The all-name burst produces +13.54 bp per review at 0 bp cost, with a 1.28 event Sharpe. A 10 bp haircut reduces the mean to +3.73 bp and the Sharpe to 0.35; its block confidence interval crosses zero. The impact top-10 portfolio earns +12.58 bp per scheduled review and +19.24 bp per active review, with a smoother 0 bp event-step path and a 2.00 active-trade Sharpe, but this descriptive improvement is selected from a strategy family and cannot be interpreted as independent confirmation.

5.2 Event-time anatomy

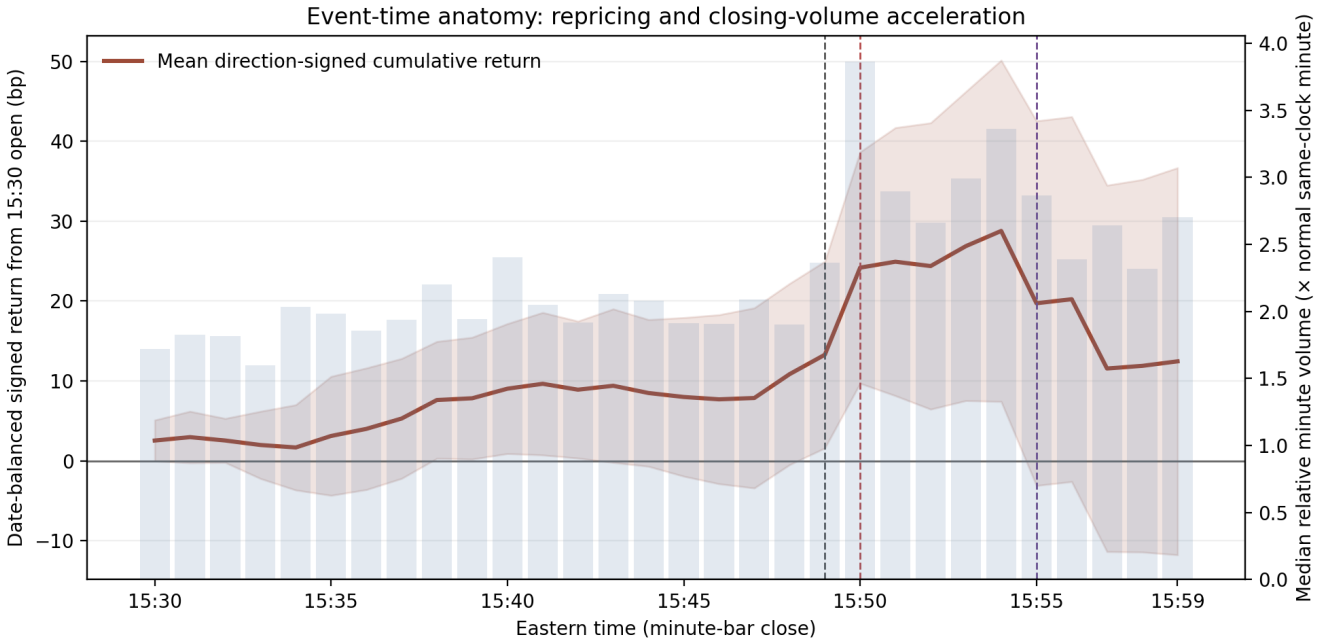


Figure 1: Direction-signed cumulative return and same-clock volume profile across 26 implementation dates. The bar timestamp denotes the minute bucket; vertical markers identify the 15:49/15:50 boundary and venue timing landmarks. Shading is descriptive across review dates. Source: audited minute-OHLCV research output.

The mean direction-signed path accelerates sharply around the 15:50 boundary, while same-clock volume is already elevated. The event profile is consistent with residual benchmark demand and imbalance information being incorporated into continuous-market prices. It is not a causal NOII test: a one-minute candle aggregates all prints in the interval and does not identify which venue message arrived first.

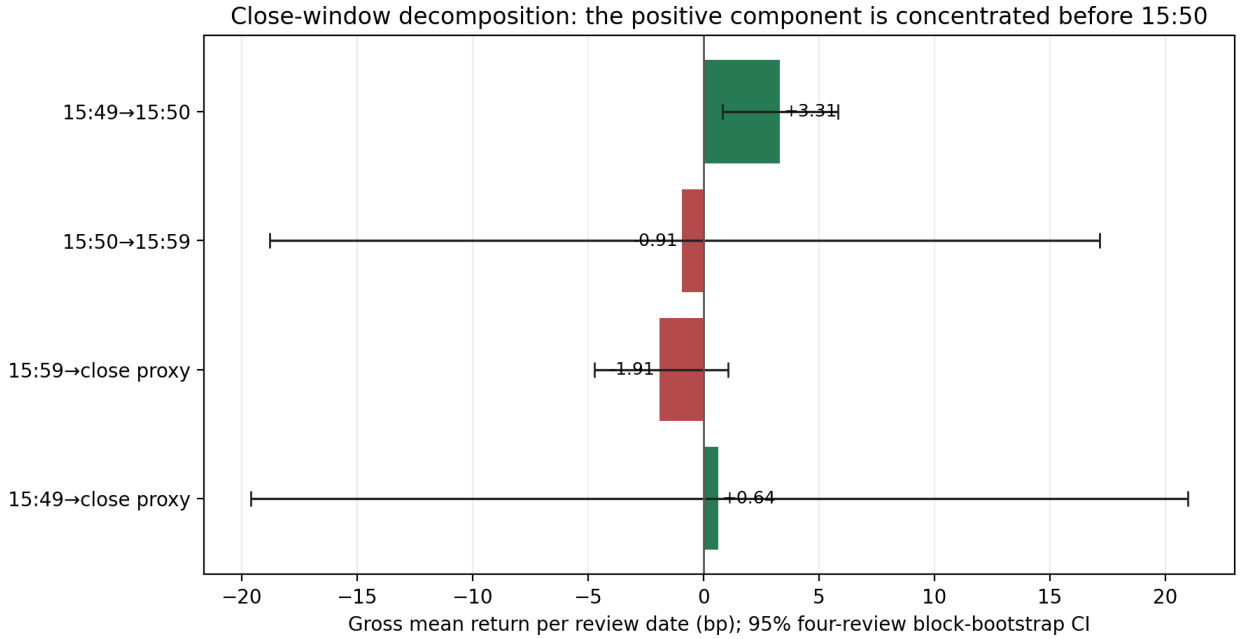


Figure 2: Non-overlapping closing-window return decomposition. Points are gross date-balanced means and bars are 95% four-review block-bootstrap intervals. The 15:49-open to 15:50-open component is positive; later continuous-market and daily-close-proxy components are unstable. Source: audited backtest output.

5.3 Equity path and cost sensitivity

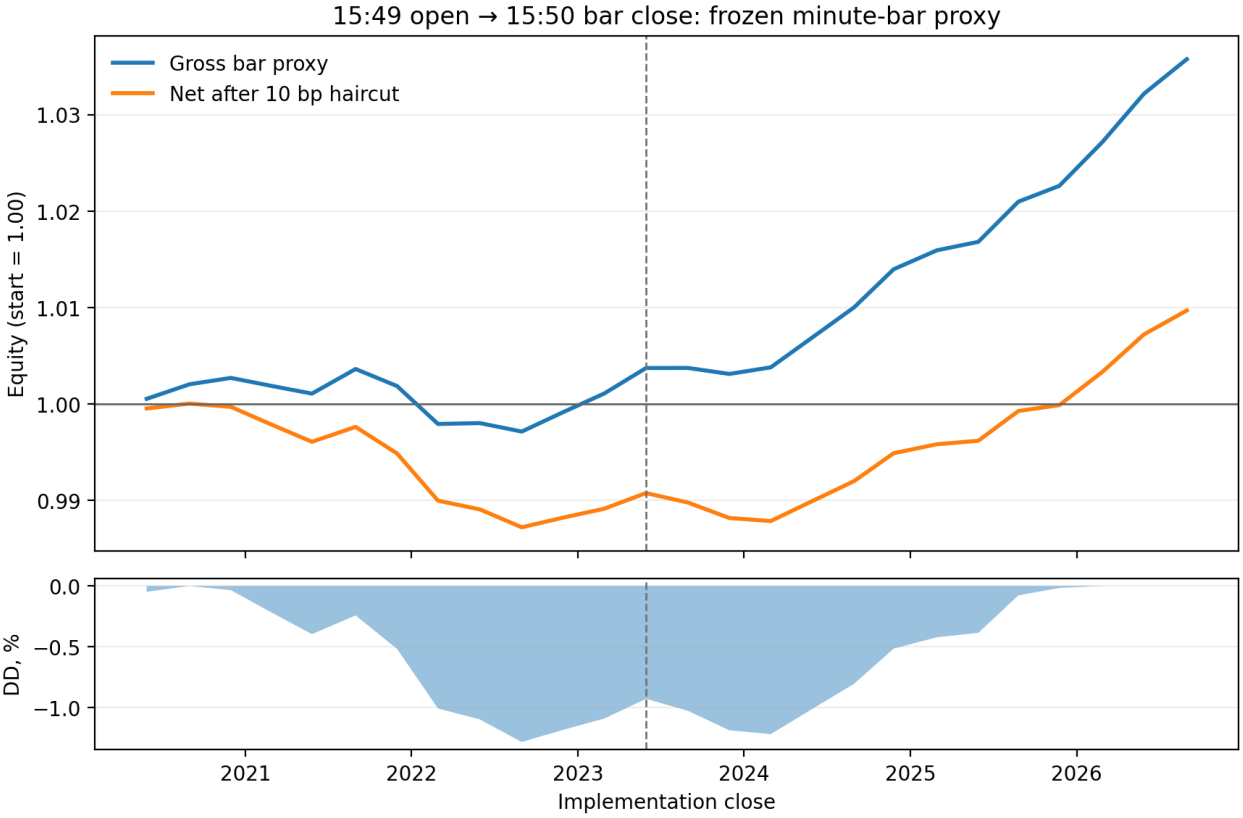


Figure 3: Event-step equity and drawdown for the all-name 15:49-open to 15:50-bar-close proxy. The gross line uses 0 bp; the conservative comparison applies the stated 10 bp haircut. Cash is held between quarterly implementation dates. Source: audited backtest output.

The gross break-even estimate is 13.80 bp per round trip. That number is an arithmetic threshold under the bar-fill proxy, not a capacity estimate. The 10 bp sensitivity remains positive in cumulative terms, but its uncertainty is wide and the realized drawdown is materially larger. Observed commissions of zero do not eliminate spread, latency, market impact, or adverse selection.

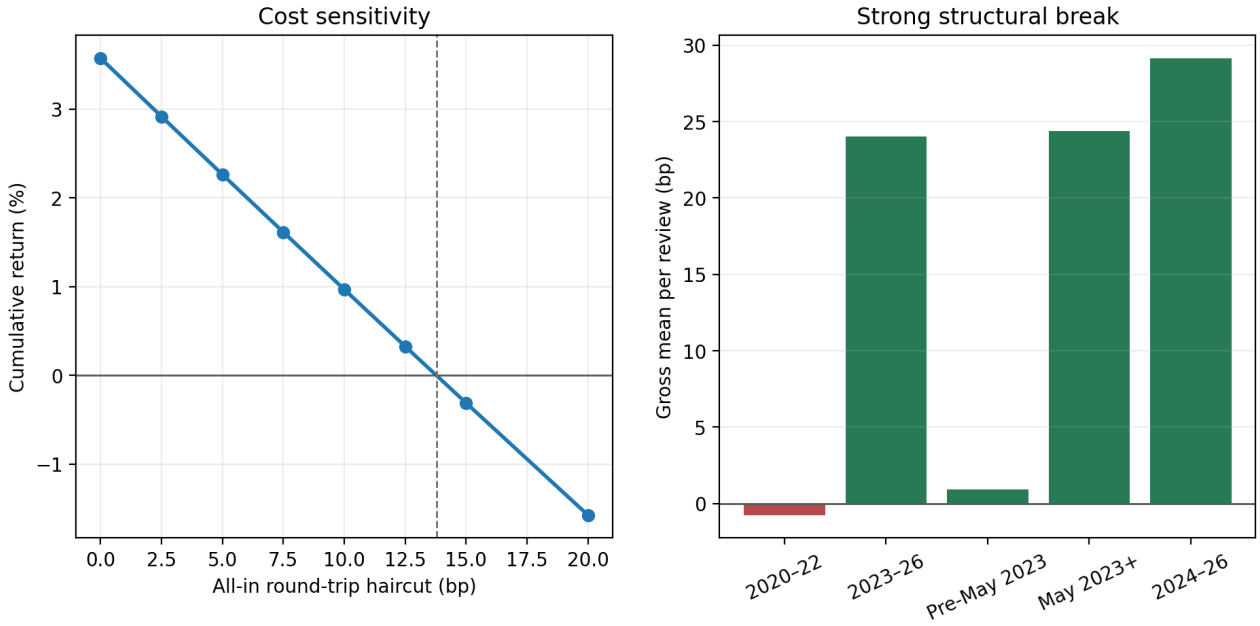


Figure 4: Left: cumulative return over an all-in haircut grid. Right: gross mean by historical slice. The vertical line marks estimated arithmetic break-even. Performance is concentrated after 2023, near the transition to comprehensive quarterly index reviews. Source: audited backtest output.

5.4 Regime dependence and discovery bias

Table 4: Regime and discovery-sample diagnostics.

Slice	Dates	Cost	Mean / review	Cumulative	Event Sharpe	Block mean CI
2020-2022	11	0 bp	-0.76 bp	-0.09%	-0.08	[-10.93, +8.16] bp
2023-2026	15	0 bp	+24.03 bp	+3.66%	2.88	[+13.83, +33.42] bp
Excluding August 2026	25	0 bp	+12.69 bp	+3.22%	1.20	[+1.82, +23.30] bp
Impact top-10, 2020-2022	11	0 bp	+5.72 bp	-	-	[-0.94, +13.87] bp
Impact top-10, 2023+	15	0 bp	+17.61 bp	-	-	[+5.40, +32.87] bp

The all-name mean is -0.76 bp in 2020-2022 and +24.03 bp in 2023-2026. Excluding the August 2026 observation that motivated the research leaves +12.69 bp per review, but does not create a genuine out-of-sample test. MSCI's first comprehensive quarterly review was implemented at the February 2023 close; full quarterly NOS/FIF review followed in May 2023 [16-18]. The first future reviews, using a frozen rule and unchanged data pipeline, are the relevant holdouts. Appendices C-E decompose every review into addition-long and deletion-short contributions and audit the 2020-2022 regime break.

5.5 Window search and controls

Table 5: Selection-adjusted and same-name control evidence; clock row is 15:49-open to 15:50-bar-close.

Test	Estimate	Interval / adjustment	Decision
Clock panel (n=443)	+13.45 bp	29-window block max-t p=0.164	No reject; n=459: +13.54 bp
Event on complete-control sample	+12.27 bp	block CI [+0.17, +23.97]	Positive event mean
Four-prior-session mean	+1.87 bp	block CI [+1.18, +2.54]	Positive but much smaller
Event minus prior-session control	+10.40 bp	block CI [-1.55, +22.33]	Interval crosses zero

Test	Estimate	Interval / adjustment	Decision
Impact top-10 lift vs matched full basket	+2.26 bp	15-rule block max-t p=0.352	Selection lift not validated

6 Identification and Robustness

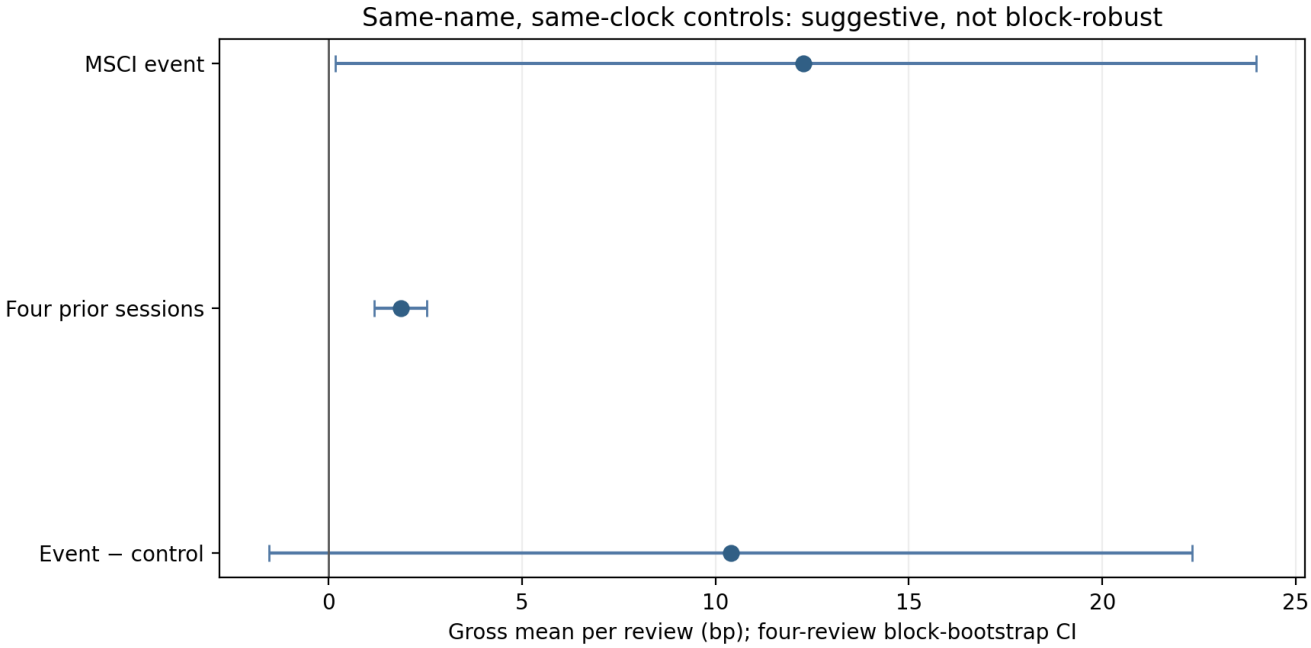


Figure 5: Same-name, same-clock comparison on 24 review dates with four complete prior sessions. Points are date-balanced means; intervals are four-review block bootstraps. The event-minus-control point estimate is positive, but its interval crosses zero. Source: audited control output.

The same-name control reduces static differences in volatility and liquidity, but it is not a complete causal design. Prior sessions do not share the review day's market shock, month-end flow, earnings news, sector news, or other index events. A stronger design would combine exact historical NOII, message-level quotes, official auction prints, contemporaneous non-event controls, and predetermined interaction terms based on licensed expected flow.

The cross-section cannot manufacture statistical breadth. Treating 459 security-events as independent would severely understate uncertainty because all names on a date share the same close, index review, and market state. All reported confidence intervals and p-values therefore operate on the 26-date series or on date-level paired differences.

7 Exploratory Top-10 Portfolio

At the selected 0 bp cost, impact top-10 earns +12.58 bp per scheduled review and +19.24 bp per active review. It compounds to +3.32%, with 0.52% cash-funded CAGR, a 2.00 active-trade Sharpe, and -0.09% event-step maximum drawdown. The 10 bp sensitivity compounds to +1.70% and has a 1.05 active-trade Sharpe.

The gross lift over the same-gate capped universe is only +2.26 bp per scheduled review, with a block interval of [-0.89, +6.03] bp and a 15-rule familywise p-value of 0.352. Removing SNDK without renormalizing lowers the gross mean from +12.58 to +11.51 bp. The motivating name matters, but it does not explain the full result.

The long and short legs receive symmetric capital budgets on active reviews, but their realized arithmetic profit shares are 59.05% from addition longs and 40.95% from deletion shorts. Before 2023, the long leg contributes 72.89 bp and the short leg contributes -9.95 bp; after the February 2023 breakpoint the figures are 120.23 and 143.87 bp. This distinction matters because signed contribution shares can exceed 100% or become negative whenever one leg offsets the other.

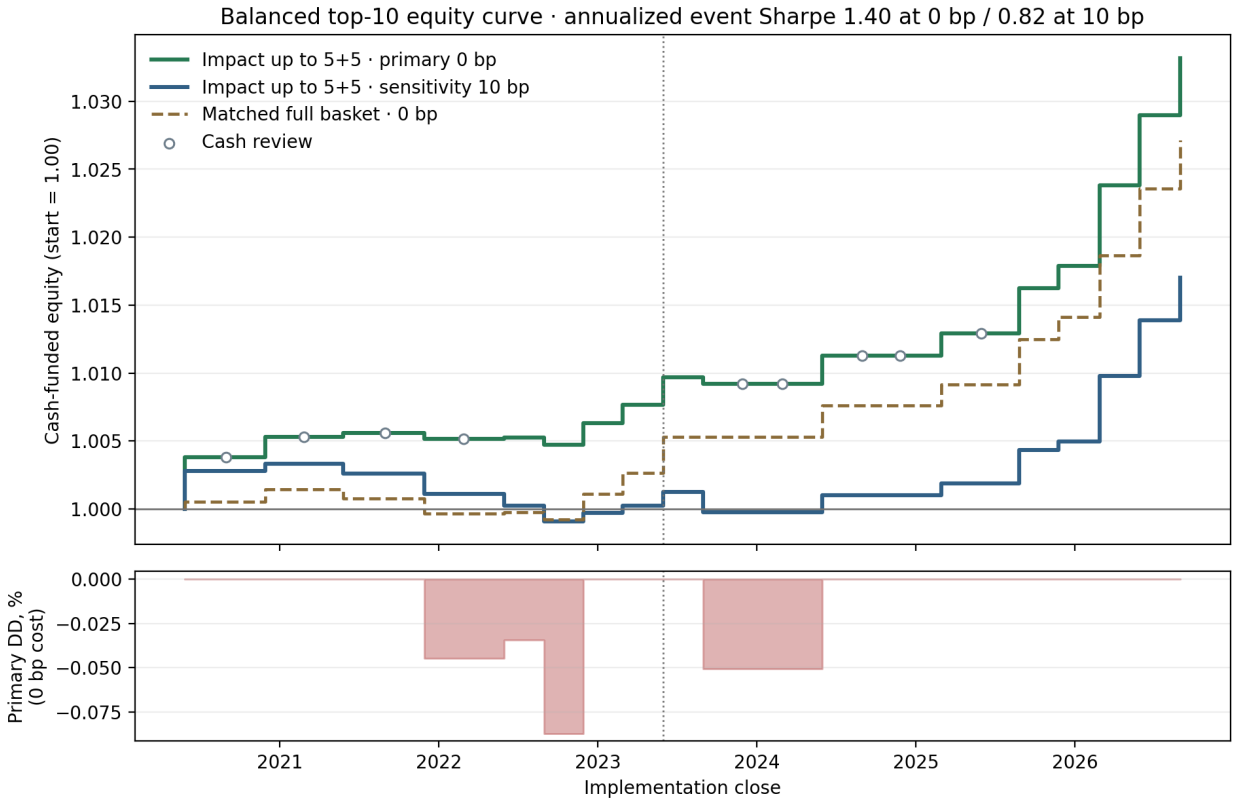


Figure 6: Exploratory impact top-10 event-step equity. The green line is the selected 0 bp primary scenario; the blue line is the 10 bp sensitivity; the dashed line is the same-gate full-basket control at 0 bp. Hollow markers denote cash reviews. The lower panel is 0 bp primary drawdown. Source: audited top-10 output.

8 Volatility Evidence and Option Boundary

Implementation names exhibit much larger pre-close movement than the same names in four prior sessions. The mean absolute terminal move is +64.5 bp on event dates versus +29.9 bp in controls, an excess of +34.6 bp with a block interval of [+22.6, +46.7] bp. This supports the premise that the closing window is unusually volatile.

The result is not an option-straddle backtest. A tradeable long-volatility claim requires historical OPRA bid, ask, size, trades, contract specifications, implied volatility, Greeks, exercise rules, and exact entry/exit timestamps. Evidence from scheduled earnings announcements shows that large underlying moves can coexist with negative delta-neutral straddle returns when event risk is richly priced [11]. The current option calculations are sensitivity analysis and remain outside the validated strategy.

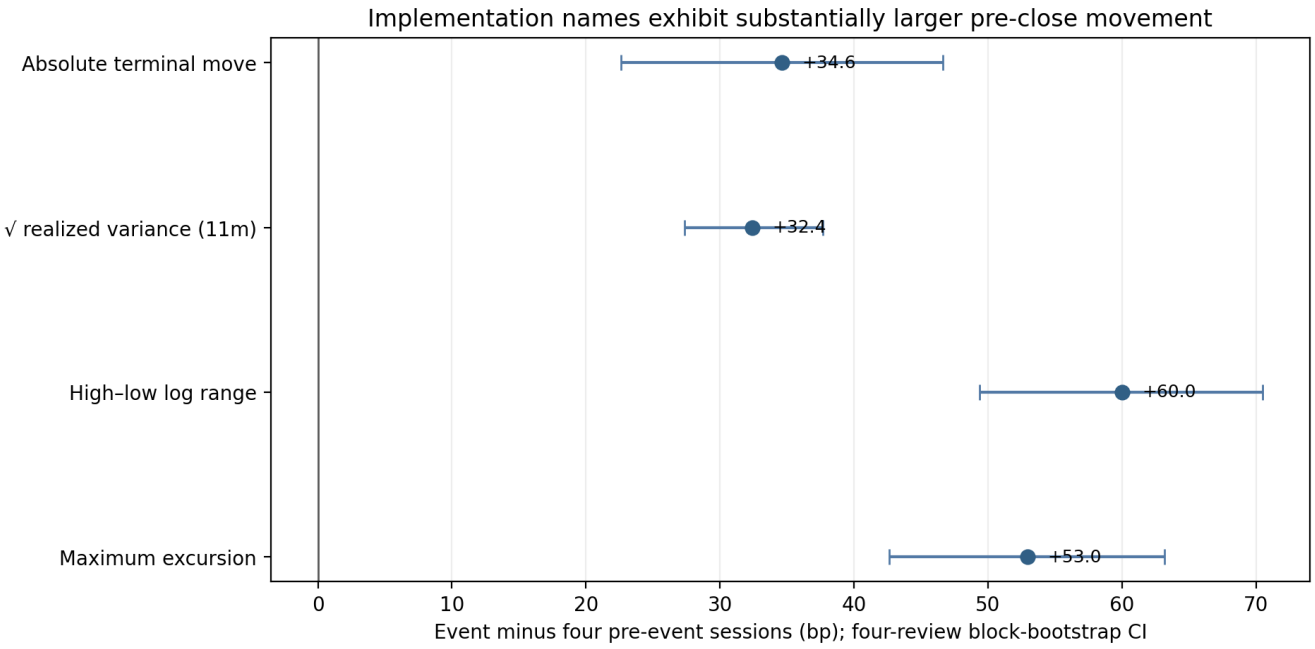


Figure 7: Event-minus-control movement diagnostics on 24 controlled review dates and 398 security-events. Intervals are four-review block bootstraps. These are underlying-price diagnostics, not option P&L. Source: audited volatility output.

9 Economic Interpretation

9.1 Why a burst may exist

A plausible chain is: announced weight change creates expected benchmark demand; residual orders accumulate toward the close; venue imbalance messages reveal a portion of the remaining demand; continuous-market liquidity providers update prices; and the official auction clears a large quantity near the newly discovered price. This mechanism explains how the largest volume can occur at or after 15:51 while much of the price change begins around 15:50. The current data show timing consistency, not message-level causality.

9.2 Why concentration may appear to help

Price impact should be larger when a given order is large relative to available liquidity and when short-horizon volatility is high [5]. The impact proxy encodes that intuition. Its descriptive improvement can also arise from selection noise, omitted actual flow, or late-sample concentration. Because the 15-rule familywise test does not reject, the score is a frozen challenger rather than a validated production rank.

9.3 Why 0 bp is informative but insufficient

The user-specified 0 bp scenario isolates the underlying price pattern. It is useful for measuring the economic signal before explicit costs. It does not imply costless execution. Bid-ask spread, price impact, queue priority, latency, borrow, rejects, and partial fills remain unobserved. The 10 bp sensitivity demonstrates that a modest haircut materially reduces performance, especially before 2023.

9.4 Pre-entry market-state and migration filters

A second-stage filter study uses only bars completed by 15:47:59.999 Eastern Time, leaving one full minute before the 15:49 entry proxy. Candidate variables include stock day, hour and week returns; side-signed stock-minus-SPY returns; intraday realized volatility; cumulative volume relative to ADV; same-clock historical RVOL; and MSCI migration type. Market-level intraday momentum and stock-level same-clock/reversal research motivate testing both continuation and exhaustion rather than imposing one sign [26-27]. One TPL row on August 31, 2022 lacks a complete safe-cutoff hour and fails closed.

The most useful mechanism rule targets the weak deletion leg. It retains every addition and true Standard exit, but requires a down-to-small deletion to have positive direction-signed stock-minus-SPY return from the 14:48 bar open through the 15:47 bar close. Surviving positions keep the original 10% cap; only the larger leg is scaled down to restore dollar neutrality, and unused capital remains cash.

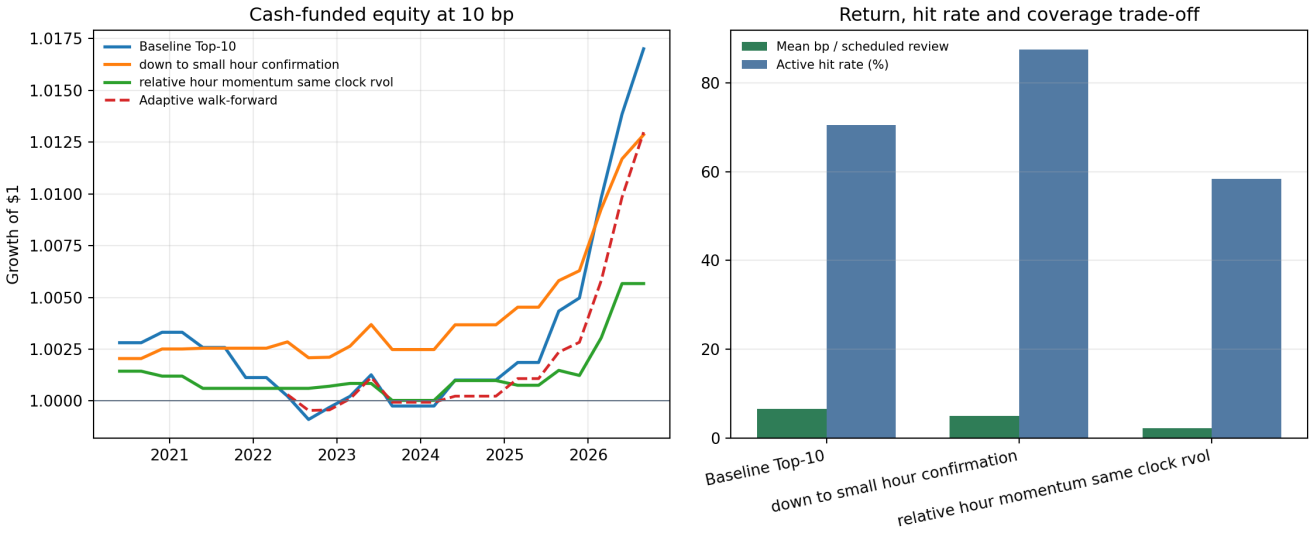


Figure 8: Cash-funded 10 bp equity for the baseline, migration-aware hour confirmation, same-clock RVOL confirmation, and expanding rule selection. The right panel shows the return, active-date hit-rate and exposure trade-off. Source: frozen 15:47 state-feature panel and review-date portfolio outputs.

Table 6: Pre-15:49 state-filter return, hit-rate and exposure trade-off at 10 bp.

Rule / sample	Reviews	Net bp	Active	Active hit	Mean gross	Active Sharpe	Max DD
Baseline / all	26	169.04	17	70.6%	60.8%	1.05	-0.42%
Migration gate / all	26	127.90	16	87.5%	38.6%	1.48	-0.12%
80% gate blend / all	26	136.13	17	76.5%	43.0%	1.32	-0.13%
Adaptive / pseudo-OOS	18	129.20	13	84.6%	45.3%	1.30	-0.12%
Baseline / matched pseudo-OOS	18	157.71	13	76.9%	65.6%	1.26	-0.20%

At 10 bp, the static migration gate raises active-date hit rate from 70.6% to 87.5% and reduces maximum drawdown from -0.42% to -0.12%. Arithmetic net profit falls from 169.04 to 127.90 bp. The paired incremental mean is -1.58 bp per scheduled review and the seven-rule block max-t p-value is 0.875; the rule does not beat baseline return.

The 80% gate blend retains 80.5% of baseline net profit and raises active-date hit rate by 5.9 percentage points. Its mixing weight was selected on the same sample and is reported only as a return-reliability frontier, not as a frozen production parameter.

The expanding selector starts after eight reviews and uses only earlier dates at every step. Across 18 pseudo-out-of-sample reviews it increases active-date hit rate from 76.9% to 84.6% while retaining 81.9% of matched-baseline net profit. This is retrospective pseudo-OOS because the candidate family was formed after the full history existed. High same-clock RVOL underperforms the low-RVOL half and is rejected as a standalone gate. The migration rule is therefore a stability challenger that requires eight genuinely future quarterly reviews.

9.5 Bounded soft-gate optimization

Before running the extension, the candidate family was frozen to the baseline, three monotone soft gates that retain 75%, 50%, or 25% of an unconfirmed down-to-small short, and the existing hard gate. Day/hour OR and AND rules are mechanism diagnostics and cannot be selected. The expanding selector begins after eight reviews and chooses the least intervention that retains at least 80% of training net profit, improves active-date hit rate by five percentage points, preserves at least 35% of baseline exposure, and has three active dates.

The 25% soft gate is the sole full-history constrained choice. After attenuation, only the larger surviving long or short leg is scaled down to restore dollar neutrality; unused capital stays in cash. This construction prevents an adverse state signal from reversing the MSCI direction and keeps the 10% single-name cap.

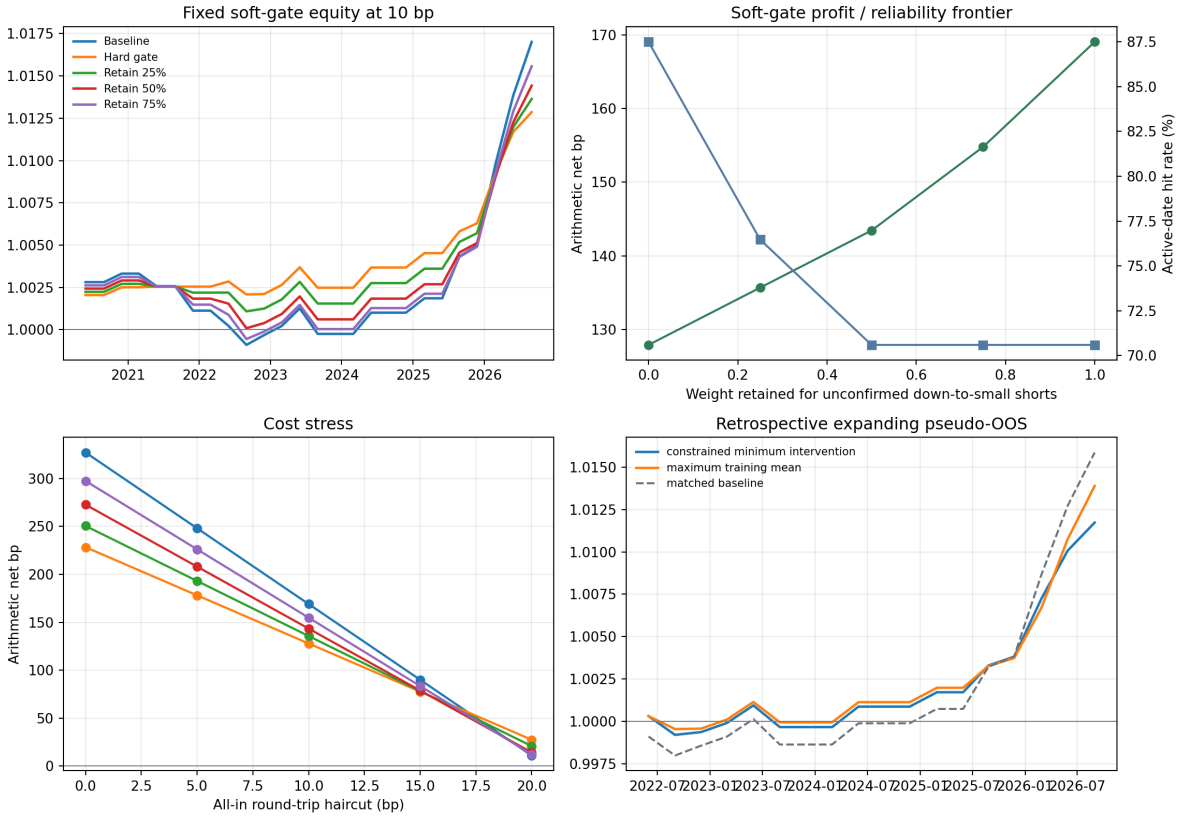


Figure 9: Frozen soft-gate optimization. Top left: fixed 10 bp equity. Top right: profit versus active-date hit rate. Bottom left: 0-20 bp cost stress. Bottom right: expanding pseudo-out-of-sample paths against the matched baseline. Source: frozen optimization specification and review-date grouped portfolios.

Table 7: Frozen soft-gate fixed and expanding pseudo-out-of-sample performance at 10 bp.

Rule / policy	Reviews	Net bp	Net retained	Active hit	Active Sharpe	Max DD	Mean gross
Baseline / full	26	169.04	100.0%	70.6%	1.05	-0.42%	60.8%
Retain 25% / full	26	135.67	80.3%	76.5%	1.25	-0.16%	44.2%
Hard gate / full	26	127.90	75.7%	87.5%	1.48	-0.12%	38.6%
Matched baseline / pseudo-OOS	18	157.71	100.0%	76.9%	1.26	-0.20%	65.6%
Constrained / pseudo-OOS	18	116.75	74.0%	84.6%	1.36	-0.13%	48.2%
Max mean / pseudo-OOS	18	138.18	87.6%	84.6%	1.41	-0.12%	48.0%

At 10 bp the 25% soft gate retains 80.3% of full-sample net profit, raises active hit from 70.6% to 76.5%, and lowers maximum drawdown from -0.42% to -0.16%. Its paired return increment is -1.28 bp per review, block CI [-4.84, 1.83], and max-t p=0.867. It improves reliability by reducing exposure, not by increasing absolute profit.

In the 18-review expanding path, the constrained policy retains only 74.0%, below its frozen 80% target. Its paired increment is -2.28 bp per review with max-t p=0.875. Leave-one-review-out selection chooses the 25% rule 69.2% of the time. An independent reconstruction of 161 security rows reconciles all 26 dates to maximum error 1.10e-16. The execution label remains minute_bar_proxy_not_broker_fill.

10 Limitations and Required Validation

- **Small independent sample.** The study has 26 review dates. Stock-level rows share common shocks and cannot be treated as independent observations.
- **Post-hoc discovery.** The interval, ranking family, original state-filter family, and 80% blend were explored on the same sample. The later monotone soft-gate family was frozen before its first run, but the full 2020-2026 history was already visible. Block-adjusted familywise tests do not reject a return improvement at conventional levels, and the constrained pseudo-OOS path misses its 80% profit-retention target.
- **Regime concentration.** Most profits occur after 2023. The February 2023 comprehensive-quarterly-review transition is both a possible mechanism and a structural-break risk.
- **No future holdout.** August 2026 motivated the research. Removing it is a sensitivity check, not a genuine out-of-sample test.
- **Proxy execution.** Minute OHLCV endpoints contain no NBBO spread, depth, venue, queue, latency, rejects, or partial fills. The official auction print is not identified.
- **Incomplete forced-flow model.** Public lists omit old/new weights, NOS, FIF, product AUM, replication fractions, crossing, and cross-index netting. Standard-Small migrations can create offsetting benchmark flows.
- **Borrow and capacity.** The short leg has no historical locate/fee panel. The current capacity score is based on trailing liquidity proxies and does not model nonlinear impact.
- **Control limitations.** Four prior sessions do not control for contemporaneous earnings, market news, other index events, sector shocks, or month-end flows.
- **Options boundary.** No historical OPRA execution data support option P&L. Underlying movement cannot substitute for an ask-in/bid-out straddle replay.
- **Top-10 artifact provenance.** The reported capped Top-10 ledgers reconcile internally, but their exact builder was a temporary, unversioned research script and the manifest lacks complete input and runner hashes. The durable repository runner applies a different sparse-date policy and produces slightly different results. The current figures therefore require generator promotion and a fully hashed rerun before independent reproduction.
- **Licensing.** Commercial use requires licensed MSCI constituent and pro-forma data plus legal review of redistribution restrictions.

10.1 Production data gate

Table 8: Minimum production and causal-validation inputs.

Component	Required production input	Current status
Event and weights	Licensed point-in-time constituents, old/new weights, NOS, FIF, amendments, permanent IDs	Public lists only
Expected flow	Tracker AUM, replication fractions, product overlap, crossing estimates	Unavailable
Equity execution	TAQ NBBO/trades, official auction print, depth, venue rules, latency and queue replay	Minute OHLCV proxy
Closing imbalance	NYSE imbalance feed and Nasdaq ITCH/NOII messages with deterministic timestamps	Unavailable
Short implementation	Locate availability, borrow fees, recalls and fail policy	Unavailable
Option implementation	OPRA NBBO/trades/size, contract master, IV/Greeks, exercise and fees	Untested
Validation	Frozen protocol and next eight quarterly reviews as sequential holdouts	No future holdout yet

11 Conclusion

The U.S. MSCI implementation sample contains a visible direction-signed price burst around 15:50. The all-name basket earns +13.54 bp per review at the selected 0 bp cost, while the exploratory impact top-10 earns +12.58 bp with a 2.00 active-trade Sharpe and -0.09% event-step maximum drawdown. These figures are economically interesting but statistically and operationally incomplete. Window selection does not survive the block max-t threshold, the ranking lift is not familywise significant, and the effect is concentrated after 2023. Arithmetic attribution assigns 59.1% of gross profit to addition longs and 40.9% to deletion shorts. The weak early regime is primarily a deletion-leg sign failure and cross-leg cancellation, while the post-2023 sample has positive contributions from both legs.

The bounded 25% soft gate is useful as a risk-control challenger. At a 10 bp haircut it raises active-date hit rate from 70.6% to 76.5% and reduces maximum drawdown from -0.42% to -0.16%, while retaining 80.3% of baseline arithmetic net profit. The constrained expanding path retains only 74.0%, below the frozen 80% target, and its paired mean increment is -2.28 bp per review. The reliability improvement comes from lower exposure; it does not establish incremental alpha.

The correct decision is to freeze the rule and run it as a shadow strategy. Production consideration requires licensed point-in-time MSCI flows, message-level imbalance and quote replay, borrow data, realistic impact, and future sequential holdouts. Until those gates are passed, the paper documents a research candidate rather than a validated arbitrage product.

A Reproducible Implementation Checklist

1. Freeze the official review announcement, implementation date, amendments, and permanent security identifiers.
2. Resolve Standard additions/deletions together with Small Cap migrations and overlapping product families.
3. For the state filter, use only bars completed by 15:47:59.999 Eastern Time; reserve 15:48:59.999 for the strict theoretical boundary and research-only cutoff sensitivity.
4. Compute the base impact-rank liquidity and volatility only from completed pre-event sessions; compute event-day state variables only through the 15:47 safe cutoff.
5. Require at least three valid additions and three valid deletions; otherwise hold the review entirely in cash.
6. Rank each side independently; select up to five names per side; break ties deterministically by ticker.
7. Assign at most 10% of capital per name and at most 50% per leg; leave unused capacity in cash.
8. Record the 15:49 start-stamped bar open as the entry proxy and the 15:50 start-stamped bar close as the exit proxy.
9. Store gross contribution, selected transaction-cost scenario, net contribution, exposure, and fixed-notional attribution for every security.
10. Aggregate to one return per review date before inference; retain all cash reviews as zero-return observations.
11. Report event-step equity, cash-funded calendar CAGR, event-step maximum drawdown, Active-trade Sharpe sqrt(4), Scheduled Event Sharpe, block intervals, and familywise p-values.
12. Do not change the rule during the next eight quarterly sequential holdouts; log deviations before observing outcomes.

B Research Ledger and Audit Trail

The 26-row review ledger is the schedule spine. It retains every active and cash decision, eligible side counts, selected names, gross and net exposure, modeled cost, event-step equity, and drawdown. The 161-row security ledger contains date, side, side rank, ticker, MSCI change type, proxy score, target weight, bar-start timestamps, entry open, exit close, gross contribution, 0 bp modeled cost, net contribution, fixed USD 1 million attribution, feature cutoff, liquidity as-of date, fill model, information cutoff, and a point-in-time-valid flag.

Dollar columns use a fresh fixed USD 1 million notional for each review and therefore sum arithmetically. The equity curve compounds capital through time, so the two dollar totals differ slightly. The security log has no placeholder rows for nine cash reviews; the date ledger supplies the complete schedule.

Table 9: Research-ledger invariants.

Audit invariant	Verified condition
Point-in-time selection	Feature cutoff and liquidity as-of strictly precede implementation date
Information clock	Production state features end at 15:47:59.999; 15:48 is research-only sensitivity
Weight integrity	Absolute name weight <=10%; sides ranked independently; cash gate enforced
P&L reconciliation	Security contributions sum to each active review within floating-point tolerance
Schedule integrity	26 unique dates: 17 traded and 9 cash
Cost integrity	Primary modeled cost exactly 0 bp; 10 bp fields retained only as sensitivity
Leg attribution	161 security rows reconcile to 26 review returns; maximum date error below 5e-16
Top-10 provenance	Frozen ledgers reconcile; exact temporary builder is not versioned and must be promoted before replication
Test status	48 automated tests pass; run archived in outputs/baseline/paper_qa_test_log.txt; execution realism remains untested

C Long and Short Profit Attribution

The active portfolio budgets up to 50% of capital to addition longs and 50% to deletion shorts. Realized arithmetic profit is less symmetric: additions contribute 193.12 bp, or 59.05% of the 327.04 bp total; deletion shorts contribute 133.92 bp, or 40.95%. These are additive contribution shares of the fixed-capital review ledger, not shares of compounded wealth.

The regime split is economically sharper. During 2020-2022, addition longs contribute 72.89 bp while deletion shorts detract 9.95 bp. The signed shares are therefore 115.8% and -15.8%. From February 2023 onward both legs are positive, with 120.23 bp from longs and 143.87 bp from shorts.

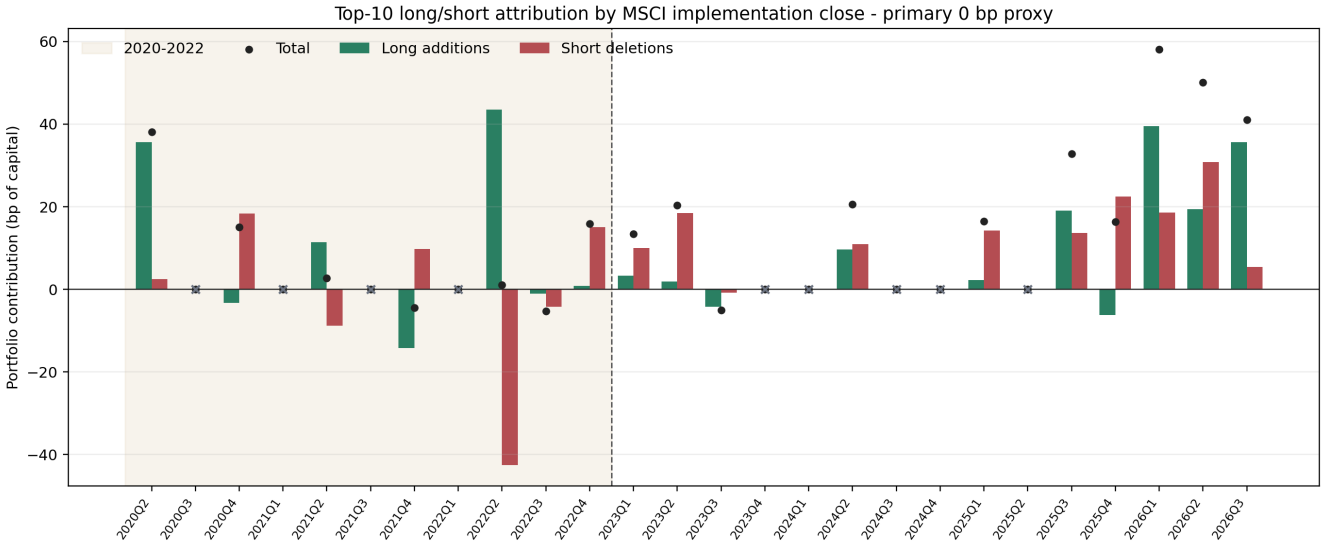


Figure 10: Addition-long and deletion-short contribution for every scheduled implementation close. Black dots are total review return; x marks are cash reviews. The shaded period is 2020-2022 and the dashed line marks the February 2023 regime boundary. Source: frozen 161-row security ledger reconciled to the 26-date schedule.

Table 10: Arithmetic leg attribution by regime.

Slice	Reviews S/A/C	Long bp	Short bp	Total bp	Signed L/S share	Both + / disagree	Net mean @10bp
All	26/17/9	+193.12	+133.92	+327.04	59.1% / 40.9%	10 / 5	+6.50 bp
2020-2022	11/7/4	+72.89	-9.95	+62.94	115.8% / -15.8%	2 / 4	-0.28 bp
2023+	15/10/5	+120.23	+143.87	+264.10	45.5% / 54.5%	8 / 1	+11.47 bp

Notes: S/A/C denotes scheduled/active/cash reviews. Signed contribution shares may exceed 100% or be negative when one leg offsets the other. The 10 bp column applies the recorded exposure-aware round-trip haircut; it is a sensitivity, not an observed fill cost.

Table 11: Calendar-quarter seasonality across the full sample.

Calendar quarter	Reviews S/A/C	Long bp	Short bp	Total bp	Mean / scheduled	Active hit L/S
Q1	6/3/3	+45.07	+42.87	+87.94	+14.66 bp	100% / 100%
Q2	7/6/1	+121.47	+11.43	+132.90	+18.99 bp	100% / 67%
Q3	7/4/3	+49.45	+14.06	+63.51	+9.07 bp	50% / 50%
Q4	6/4/2	-22.87	+65.56	+42.69	+7.12 bp	25% / 100%

Notes: Q1-Q4 pool only six or seven scheduled reviews each and have unequal active counts. These figures describe seasonality; they are not independent quarterly subsamples.

D Review-by-Review Performance Ledger

Table 11 reports all 26 nonempty analyzed reviews from May 2020 through August 2026, including nine cash decisions. The February 2020 public review had zero U.S. Standard additions and deletions and lies outside the return sample. Long and short columns are portfolio contributions in basis points of total capital. The eligible counts are the point-in-time-valid additions and deletions before the three-per-side gate; selected counts are shown separately.

Table 12: All 26 nonempty analyzed MSCI implementation closes in the Top-10 ledger.

Quarter	Date	Status	n L/S	Long	Short	Gross	Net10	Abs L/S	Elig A/D	Largest absolute name
2020Q2	2020-05-29	Traded	5/5	+35.58	+2.54	+38.12	+28.12	93%/7%	23/41	MRNA +19.97bp
2020Q3	2020-08-31	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	3/1	-
2020Q4	2020-11-30	Traded	5/5	-3.28	+18.30	+15.02	+5.02	15%/85%	19/14	NOV +9.74bp
2021Q1	2021-02-26	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	2/1	-
2021Q2	2021-05-27	Traded	5/5	+11.46	-8.79	+2.67	-7.33	57%/43%	11/5	IONS -8.00bp
2021Q3	2021-08-31	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	2/1	-
2021Q4	2021-11-30	Traded	5/5	-14.22	+9.74	-4.48	-14.48	59%/41%	11/8	SOFI -14.09bp
2022Q1	2022-02-28	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	1/1	-
2022Q2	2022-05-31	Traded	5/5	+43.57	-42.54	+1.03	-8.97	51%/49%	12/11	ASAN -13.14bp
2022Q3	2022-08-31	Traded	3/3	-1.04	-4.22	-5.27	-11.27	20%/80%	3/3	TPL +6.36bp
2022Q4	2022-11-30	Traded	5/5	+0.82	+15.03	+15.85	+5.85	5%/95%	14/12	APE +12.76bp
2023Q1	2023-02-28	Traded	5/4	+3.30	+10.06	+13.36	+5.36	25%/75%	7/4	TEAM +3.67bp
2023Q2	2023-05-31	Traded	5/5	+1.85	+18.52	+20.37	+10.37	9%/91%	8/7	NWL +7.76bp
2023Q3	2023-08-31	Traded	5/5	-4.20	-0.86	-5.05	-15.05	83%/17%	6/6	DKNG -3.62bp
2023Q4	2023-11-30	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	2/17	-
2024Q1	2024-02-29	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	4/2	-
2024Q2	2024-05-31	Traded	4/5	+9.63	+10.92	+20.55	+12.55	47%/53%	4/14	PSTG +5.84bp
2024Q3	2024-08-30	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	1/9	-
2024Q4	2024-11-25	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	2/4	-
2025Q1	2025-02-28	Traded	4/5	+2.21	+14.28	+16.48	+8.48	13%/87%	4/16	RDDT +6.05bp
2025Q2	2025-05-30	Cash	0/0	+0.00	+0.00	+0.00	+0.00	-	0/28	-
2025Q3	2025-08-26	Traded	4/4	+19.07	+13.70	+32.77	+24.77	58%/42%	4/4	RKLB +7.97bp
2025Q4	2025-11-24	Traded	5/5	-6.20	+22.50	+16.30	+6.30	22%/78%	12/14	OKLO -9.55bp
2026Q1	2026-02-27	Traded	5/5	+39.56	+18.53	+58.09	+48.09	68%/32%	8/15	ASTS +11.98bp
2026Q2	2026-05-29	Traded	5/5	+19.38	+30.77	+50.15	+40.15	39%/61%	7/19	CRCL +8.17bp
2026Q3	2026-08-31	Traded	5/5	+35.63	+5.44	+41.06	+31.06	87%/13%	6/18	SNDK +27.70bp

Notes: Gross = Long + Short. Net10 applies the 10 bp exposure-aware sensitivity. Abs L/S divides the absolute leg contributions on that date and is undefined for cash reviews. Largest absolute name may be a gain or loss. Minute-bar endpoints remain research fill proxies.

E Why the 2020-2022 Effect Is Weak

The early Top-10 estimate is weak rather than literally zero: 5.72 bp per scheduled review across 11 quarters, with only seven active reviews and a four-review block interval of [-0.94, +13.87] bp. A 10 bp haircut changes the mean to -0.28 bp. Four cash reviews and cross-leg cancellation therefore consume most of the gross signal.

Weak signed return does not imply missing mechanical flow. NYSE reports that the May 29, 2020 MSCI rebalance closed 1.55 billion shares and approximately USD 60 billion, 21.2% of that day's NYSE-listed volume [20]. Removing that review lowers rather than improves the early result, so May 2020 actually supports the early estimate.

The central accounting failure is the deletion leg. Addition longs earn 72.89 bp, deletion shorts lose 9.95 bp, and the active-date leg correlation is -0.733. On 2022-05-31 alone, longs add +43.57 bp while shorts lose 42.54 bp, leaving only +1.03 bp. The two legs are simultaneously positive on two of seven active early dates, versus eight of ten after the February 2023 breakpoint.

Migration mechanics give a plausible reason for the sign failure. In the all-name early sample, 95 of 99 deletions are down-to-small migrations, and 66 of 105 additions are up-from-small migrations. Standard-index selling can be offset by Small Cap tracker buying, so a public Standard add/delete label is not a net-flow estimate. After first averaging within each review date, early down-to-small deletion shorts average -6.26 bp, versus +21.12 bp after 2023; new-to-Standard additions average +8.13 versus +24.85 bp, and up-from-small additions +1.34 versus +25.59 bp. These pooled-regime summaries are descriptive and have no formal clustered breakpoint test. Testing the mechanism requires old/new weights, NOS, FIF, Standard and Small Cap tracker AUM, and cross-index netting.

MSCI implemented its first comprehensive quarterly index review at the February 28, 2023 close and began full quarterly NOS/FIF reviews in May 2023 [16-18]. February/August coverage expands from 3.6 to 14.25 all-name events per review in this sample. Comparable May/November reviews also improve, so QCIR explains sparse early quarters and a change in event composition, but it does not by itself explain the full return break.

The exact 15:50 signed minute changes from -0.66 bp in 2020-2022 to 18.72 bp after the breakpoint; mean 15:50 volume share rises from 1.69% to 2.48%. This is consistent with more concentrated price discovery, but minute OHLCV cannot identify NOII causality [21]. Nasdaq moved its NOII start to 15:50 in 2018, and NYSE's D-order feed inclusion moved to 15:50 in August 2024. NYSE also introduced new floor-broker connectivity in late 2022 and subsequently reported a higher D-order share [22]. We find no common NYSE/Nasdaq 2023 imbalance-dissemination-time change, but a venue-composition channel cannot be excluded without venue splits [19,22,23].

One-off exclusions refine the conclusion. Removing May 2020 lowers the early Top-10 mean to +2.48 bp and the all-name mean to -1.36 bp; removing May 2023, when MSCI also implemented one-time GICS revisions [24], leaves +17.41 and +23.85 bp, respectively. Removing all 2026 observations leaves the broad all-name mean at 19.38 bp, but compresses the Top-10 mean to 9.57 bp, close to the early +5.72 bp estimate. Moreover, 2026 supplies 45.7% of total Top-10 arithmetic profit. The all-name regime pattern is more persistent; the concentrated Top-10 result is recent and needs future holdout evidence.

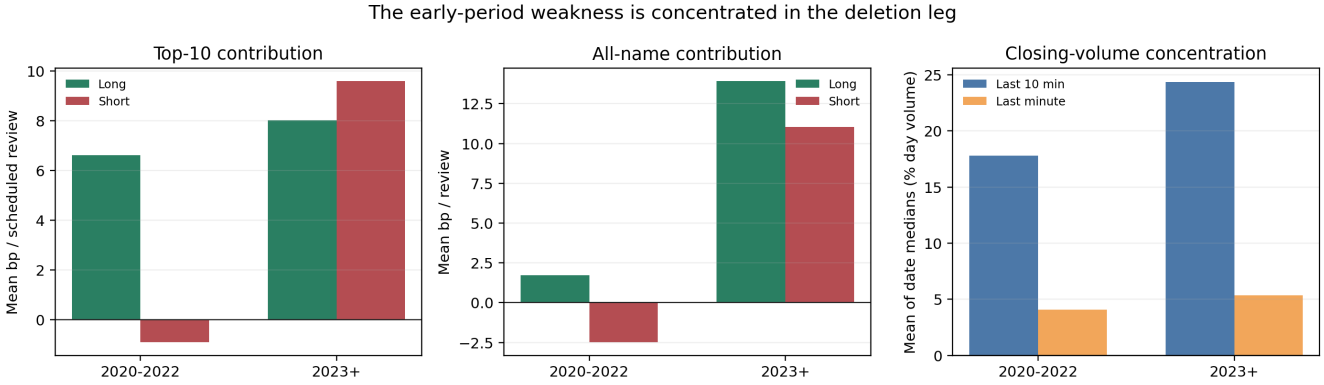


Figure 11: Regime decomposition. Left: Top-10 long and short mean contribution per scheduled review. Center: the same decomposition for the all-name basket. Right: closing-volume concentration. The deletion leg, rather than the addition leg, accounts for most of the early-to-late return change. Source: audited portfolio and volume outputs.

Table 13: Observed drivers of the early/late difference.

Diagnostic	2020-22	2023+	Difference	Unit	Interpretation
Top-10 mean gross return per scheduled review	5.72	17.61	+11.88	bp	Weak early total; materially larger late-period estimate
Top-10 long contribution per scheduled review	6.63	8.02	+1.39	bp	Long additions are positive in both periods
Top-10 short contribution per scheduled review	-0.90	9.59	+10.50	bp	Deletion short leg is the main early-period failure
Top-10 leg netting capture	29.81	92.15	+62.34	%	Early legs frequently offset; late legs usually align
All-name mean gross return per review	-0.76	24.03	+24.79	bp	Regime break is not unique to Top-10 selection
All-name February/August events per review	3.60	14.25	+10.65	names	QCIR substantially broadened the formerly light reviews
Mean median last-10-minute volume share	17.79	24.37	+6.59	% day volume	Closing volume became more concentrated after 2023
Mean median last-minute volume share	4.09	5.34	+1.25	% day volume	Terminal-minute concentration is also higher late

Table 14: Date-balanced all-name signed return by migration type.

Regime	Side	Change type	Dates	Names	Mean date bp	Date hit
2020-2022	Long	new_to_standard	9	39	+8.13	55.6%
2020-2022	Long	up_from_small	11	66	+1.34	45.5%
2020-2022	Short	down_to_small	11	95	-6.26	54.5%
2020-2022	Short	exit_standard	2	4	-2.45	50.0%
2023+	Long	new_to_standard	12	18	+24.85	75.0%
2023+	Long	up_from_small	13	58	+25.59	84.6%
2023+	Short	down_to_small	15	174	+21.12	73.3%
2023+	Short	exit_standard	4	5	+52.36	100.0%

Table 15: All-name return by review class before and after QCIR.

Regime	Review class	Dates	Names/date	Long bp	Short bp	Gross bp
2020-2022	February/August	5	3.60	+0.63	-3.75	-3.12
2020-2022	May/November	6	31.00	+2.64	-1.45	+1.20
2023+	February/August	8	14.25	+15.78	+8.79	+24.56
2023+	May/November	7	20.14	+11.43	+13.62	+23.41

Interpretation boundary: The data document a QCIR-aligned descriptive return break, a date-balanced migration-leg sign change, and stronger 15:50 concentration after 2023. No formal clustered breakpoint test is claimed, and actual net forced flow is unidentified. Causal validation requires licensed delta weights and tracker AUM plus message-level NOII, NBBO, venue and auction data.

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